ATLANTIC ANALYTICAL LABORATORY

Product Combustion Analysis Report

The Aeolus Group LLC
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Attn: Mr. Rudy Behrens
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AAL Number: 2827
Received On: 03/15/07
Report Date: 03/23/07
PO Number: Check

Sample ID.: BioDiesel Fuel Sample

Comments: 1 sample received for product combustion characterization. This material was subjected to controlled thermooxidative stress in Air. Gaseous effluents were analyzed by Gas Chromatography (GC-TCD/FID and GC-SCD) and Gas Chromatography/Mass Spectrometry (GC/MS) methods as requested.

Experimental Conditions

Air Carrier at 700° F. Gas Sample collected with furnace ramping from room temperature to desired temperature. The sample is then cooled and total weight loss determined.

Test Condition Summary

Sample wt/run = 0.5564g
Sample Boat: Porcelain
Gas Type: Air
Gas Flow = 200 cc/min.
Effluent Collection Period = 20 min.
Effluent Gas Collection Media = 5L Tedlar Gas Bag
Analytical Methodologies: GC-TCD/FID, GC-SCD, GC/MS
Ash Residue after Combustion Analysis: 0.09% w/w
Sample Weight Loss after Combustion Analysis: 99.01% w/w

Comments:

Report Key:

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GC-TCD/FID RESULTS

Composition

Non-Condensable Gases (% v/v by GC)	Result	D.L.
Nitrogen:	71.5	0.01
Oxygen:	19.7	0.01
Carbon Dioxide:	0.68	0.01
Carbon Monoxide:	0.64	0.01
Hydrogen:	0.34	0.01
Argon:	nd	0.01
Methane:	1.01	0.001

Hydrocarbons (% v/v by GC)	Result	D.L.
Ethylene:	0.698	0.001
Ethane:	0.362	0.001
Propylene:	0.431	0.001
Propane:	0.034	0.001
Isobutane:	0.140	0.001
n-Butane:	0.169	0.001
Butene:	nd	0.001
Isopentane:	0.059	0.001
n-Pentane:	0.016	0.001
Pentenes:	nd	0.001

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GC/MS RESULTS

Libr. Hit List &/or Target Analyte(s)	Total Amount Recovered (mg)	% Recovered (w/w)
Methanol	0.8	0.15
Acetaldehyde	0.7	0.13
Acrolein	1.8	0.33
Methyl acetate	3.4	0.61
Methylpentene	8.9	1.6
Cyclopentadiene	13	2.4
Methylcyclopentadienes	6.1	1.1
Cyclohexadiene	3.8	0.69
Benzene	25	4.4
C3-benzenes	0.8	0.14
C4-benzenes	0.2	0.04
C5-benzenes	0.1	0.02
Toluene	9.5	1.7
Ethylbenzene	1.8	0.33
m+p-xylenes	1.2	0.22
o-xylene	0.8	0.14
Styrene	2.5	0.45
C1-styrenes	0.6	0.14
Napthalene	0.2	0.03
Methyl Esters of Mono-unsaturated Carboxylic Acids:		
C_3 acid chain	14	2.5
C_4 acid chain	5.2	0.93
C_5 acid chain	0.3	0.05
C_6 acid chain	2.4	0.43
C_7 acid chain	2.5	0.45
C_8 acid chain	0.6	0.11
C_9 acid chain	0.5	0.09
C_{10} acid chain	0.7	0.12
C_{11} acid chain	0.3	0.05

Comments:

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GC-SCD RESULTS

Volatile Sulfur Compounds (VSC, ppb v/v)	Result	D.L.
Hydrogen Sulfide (H_2S):	28	20
Carbonyl Sulfide (COS):	100	20
Sulfur Dioxide (SO_2):	nd	20
Methyl Mercaptan:	nd	20
Ethyl Mercaptan:	nd	20
Dimethyl Sulfide:	nd	20
Carbon Disulfide:	nd	20
Isopropyl Mercaptan:	nd	20
t-Butyl Mercaptan:	nd	20
n-Propyl Mercaptan:	nd	20
Methyl Ethyl Sulfide	nd	20
sec-Butyl Mercaptan:	nd	20
Thiophene:	nd	20
Isobutyl Mercaptan:	nd	20
Diethyl Sulfide:	nd	20
n-Butyl Mercaptan:	nd	20
Dimethyl Disulfide:	nd	20
3-Methylthiophene:	nd	20
Tetrahydrothiophene:	nd	20
2-Ethylthiophene:	nd	20
2,5-Dimethylthiophene:	nd	20
Diethyl Disulfide:	nd	20
Unknown Compounds:	nd	20

Comments: Total Sulfur as S = 128 ppb v/v (0.128 ppm v/v).

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Reviewed and Approved By,

Ralph J. Ciotti

Attachments:
none

Addendum:
Chromatograms on-file
Notebook data on-file

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